Homework 8

1. A gas of atoms is confined in a cube of volume $V = L^3$. Using the formula we derived in class for the volume in k-space per wave state for standing waves, determine the total number of states available to the atoms for magnitude of k less than $2\pi/(5\text{ Å})$.

$k < \frac{2\pi}{(5\text{Å})}$, $V = L^3$

volume per state in k-space: $\frac{\pi^3}{L^3} = \left(\frac{\pi}{L}\right)^3$

standing wave k-volume: allowed volume: $\frac{1}{8}\left(\frac{4}{3}\pi k^3\right)$ (all positive)

$$\frac{\frac{1}{8}\left(\frac{4\pi}{3}k^3\right)}{\left(\frac{\pi}{L}\right)^3} = \frac{1}{8}\frac{4\pi}{3}k^3 \cdot \frac{L^3}{\pi^3} = \frac{1}{2}\frac{1}{3\pi^2}k^3L^3$$

$$\hookrightarrow \frac{k^3L^3}{6\pi^2} = \frac{\frac{(2\pi)^3}{(5\text{Å})^3}\cdot L^3}{6\pi^2} \Rightarrow \frac{L^3}{6\pi^2}\cdot\frac{8\pi^3}{125\text{Å}^3}$$

2. In class we derived the ideal gas law $PV = Nk_BT$ using a microscopic derivation for the force due to momentum of the atoms bouncing off a wall. Here you will derive this a different way in the following steps:

a) Using the result derived in class $U = \frac{3}{2}Nk_BT$, write the pressure $P = \frac{dU}{dV}$ in terms of $\frac{dT}{dV}$, assuming that N is constant.

$U = \frac{3}{2}Nk_BT$

$dU = \frac{3}{2}Nk_BdT$ and $dU = dQ - PdV$ w/ $dQ = 0$

$dU = -PdV$

$\Rightarrow$ w/ $PV = Nk_BT \rightarrow P = \frac{Nk_BT}{V}$

we get $dU = \frac{-Nk_BT}{V}dV$

and resolve: $\frac{3}{2}\cancel{Nk_B}\frac{dT}{dU} = \frac{-\cancel{Nk_B}T}{V}dV \Rightarrow \frac{3}{2}dT = -\frac{T}{V}dV \Rightarrow \frac{dT}{dV} = \frac{-2T}{3V}$

b) Using the result derived in class $N = (\text{const})V(k_BT)^{3/2}$ for an ideal gas, solve for T.

$$N = V(k_BT)^{3/2} \Rightarrow \frac{N}{V} = (k_BT)^{3/2} \Rightarrow \left(\frac{N}{V}\right)^{2/3} = k_BT \Rightarrow T = \left(\frac{N}{V}\right)^{2/3}\cdot\frac{1}{k_B}$$

c) Find $\frac{dT}{dV}$ from the result of (b), assuming that N is constant, and substitute this into your answer for (a). Show that this gives the ideal gas law and that $P = \frac{2}{3}U$.

a) $P = \frac{dU}{dV} = \frac{3}{2}Nk_B\frac{dT}{dV}$

b) $T = \left(\frac{N}{V}\right)^{2/3}\cdot\frac{1}{k_B} \leftarrow c$

$\hookrightarrow c\left(\frac{N}{V}\right)^{2/3}$

$T = cN^{2/3}V^{-2/3}$

$\frac{dT}{dV} = cN^{2/3}\frac{-2}{3}V^{-5/3}$

and since $T = cN^{2/3}V^{-2/3}$

$\frac{dT}{dV} = T\frac{-2}{3}V^{-1} = \frac{-2}{3}\frac{T}{V}$

Using $P = \frac{3}{2}Nk_B\frac{dT}{dV}$: $P = \frac{3}{2}Nk_B\frac{-2}{3}\frac{T}{V} = -Nk_B\frac{T}{V}$

and $P = \frac{-dU}{dT} = \frac{-3}{2}Nk_B\frac{dT}{dV} \Rightarrow \frac{-3}{2}Nk_B\frac{-2}{3}\frac{T}{V} = Nk_B\frac{T}{V} = \frac{Nk_BT}{V}$, thus: $PV = Nk_BT$

$\Rightarrow U = \frac{3}{2}Nk_BT$, want: $P\frac{2}{3}U \Rightarrow PV = \frac{2}{3}U$

$\hookrightarrow \frac{2}{3}U = Nk_BT$

$PV = Nk_BT$

so, $PV = \frac{2}{3}U = \frac{2U}{3V}$

3. The full formula from class for an idea gas is

$$N = V\frac{\sqrt{2}}{2\pi^2}\frac{(mk_BT)^{3/2}}{\hbar^3}\, e^{\mu/k_BT}\int_0^\infty d\varepsilon\, \varepsilon^{1/2}\, e^{-\varepsilon}$$

where the unitless integral is equal to $\sqrt{\pi}/2$.

a) Assuming that the chemical potential μ is a large negative number, if you double the number of atoms N, by what amount does μ change?

$$N = V\frac{\sqrt{2}}{2\pi^2}\frac{(mk_BT)^{3/2}}{\hbar^3}e^{\mu/k_BT}\cdot\frac{\sqrt{\pi}}{2}$$

$$\hookrightarrow \left.\begin{aligned} N &= Ce^{\mu/k_BT} \\ 2N &= Ce^{\mu'/k_BT}\end{aligned}\right\} \quad \frac{2N}{N} = \frac{Ce^{\mu/k_BT}}{Ce^{\mu'/k_BT}} \Rightarrow 2 = e^{(\mu-\mu')/k_BT}$$

Take ln: $\ln(2) = \frac{\mu-\mu'}{k_BT} \Rightarrow \mu-\mu' = k_BT\ln(2)$, so μ will have to increase

b) For a density of atoms equal to 10^{23} /cm^{-3}, each atom with a mass of 100 protons (the proton mass is given by $mc^2 = 9\text{x}10^8$ eV), at room temperature equal to 300 K, find the chemical potential.

This problem is much easier to solve if you use "natural" units of eV, not kilograms!

Boltzmann's constant: 8.6 x10^{-5} eV/K
Planck's constant times speed of light: $\hbar c = 2$ x10^{-5} eV-cm

$$= \frac{V}{(2\pi)^2}e^{\mu/k_BT}\frac{\sqrt{2}m^{3/2}(k_BT)^{3/2}}{\hbar^3}\left[\int_0^\infty \varepsilon^{1/2}e^{-\varepsilon}d\varepsilon\right] = \frac{\sqrt{\pi}}{2} = V\frac{\sqrt{2}}{2\pi^2}\frac{(mk_BT)^{3/2}}{\hbar^3}e^{\mu/k_BT}\cdot\frac{\sqrt{\pi}}{2}$$

$\rho = 10^{23}\,\text{atom}/\text{cm}^{-3}$

1 atom = 9×10^{10} eV

$E_\rho = 9\times10^{23}/\text{cm}^{-3}$

$$\boxed{\text{density of particles: } \underset{\substack{\parallel \\ n}}{\frac{N}{V}} = \frac{2\sqrt{2}}{2\pi^2}\frac{m^{3/2}}{\hbar^3}\left(\frac{2}{3}E_F^{3/2}\right)}$$